Aiden Zhang

437-788-7616 | a522zhan@uwaterloo.ca | linkedin.com/in/aidenzhangg | github.com/aidenzhangg

Technical Skills

Languages: Java, Python, C++, SQL, HTML, CSS, TypeScript, JavaScript
Frameworks: React.js, PyTorch, TensorFlow, scikit-learn, OpenCV, YOLO, ROS 2, pandas, numpy, Spring Boot
Developer Tools: Git, Github, Docker, VSCode, Foxglove, PyCharm, Eclipse, Vite, Chrome DevTools
Libraries: OpenAPI

Projects

Computer Vision Food Recognition and Nutritional Analysis System

Python, PyTorch, YOLO, OpenCV, NumPy, pandas, TensorFlow June 2025 - October 2026

- Built a real-time food detection app that finds and identifies 5 food categories in live webcam video, replacing an earlier TensorFlow classifier (96% test accuracy) with a fine-tuned YOLO11 model that boxes each food and detects several items at once averaging ~13.5 FPS across 300 frames
- Trained the model on **245 hand-labeled images, reaching 98% mAP50** on a held out validation set
- Ran a 5-month pilot with **6 peers and family members,** including a **visually impaired participant** who used TTS to access nutrition feedback supporting nutrition-aware food choices; participants reported healthier eating habits and an average **1.1% reduction in body-fat percentage**

Autonomous Robot Navigation & LiDAR Mapping - WATonomous

C++, ROS2, Foxglove, Docker Sept 2026

- Developed a 4-node autonomous navigation pipeline in C++ and ROS 2, integrating LiDAR mapping, odometry-based map fusion, A* pathfinding, and 10Hz Pure Pursuit control
- Engineered 100 x 100 m occupancy grids with 0.1m resolution, 0.65 m obstacle inflation, and 8-direction A* planning for obstacle-aware navigation

Autonomous Drone Behaviour Tree Telemetry System

Python, ROS2, MAVLINK, py_trees, pymavLink, Git Sept 2026 - Present

- Implemented an initial ROS 2 behaviour-tree telemetry protocol, subscribing to live airside snapshots and automatically mapping 100% of observed behaviors to compact IDs and parent-child relationships for bandwidth-constrained MAVLink STATUSTEXT
- Developed an air-to-ground telemetry pipeline to encode active decision paths and reconstruct autonomy state groundside, targeting ≤ 50-byte STATUSTEXT payloads for debugging without LTE connectivity

Experience

Camp Counsellor - *City of Richmond Hill* May 2025 - August 2026

- Led daily programs for children ages 4-12, working with staff and parents to resolve issues

Certificates

Stanford AI Machine Learning Specialization July 2025 - October 2025

PyCharm, PyTorch, Python, TensorFlow, Scikit-learn, NumPy, Langchain, LLM

- Built ML models using neural networks, regression, decision trees, clustering, and recommender systems

Education

University of Waterloo - *BASc in Computer Engineering with Co-op* Sept 2026 - April 2031

- **Coursework:** C++ Programming, Physics, Linear Algebra, Calculus, Engineering Prototype and Design